Generic Motion Transfer from Video to Static Image Using Deep Learning Techniques

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Abstract

This paper presents a practical motion transfer system that animates a static portrait image by transferring head motion and facial expressions from a driving video. Built on the First Order Motion Model, the system extends it with a modular compositing pipeline addressing real-world deployment challenges. The pipeline incorporates unsupervised keypoint detection, relative dense motion estimation, and Exponential Moving Average temporal smoothing to reduce inter-frame jitter. A combined masking strategy intersects a convex hull mask derived from keypoint geometry with an occlusion-aware mask from the generator's confidence output, producing feathered boundaries for seamless blending. Geometric alignment via similarity transformation positions generated frames correctly in the driving domain. Gradient-domain Poisson blending composites the animated face onto the driving background without visible seams. Optional per-channel color correction resolves appearance mismatches between source and driving domains. A best-frame alignment strategy eliminates initialization artifacts. A ConvLSTM-based temporal refinement module is under development to further reduce residual flickering.

Keywords: Image Animation, Motion Transfer, First Order Motion Model, Unsupervised Keypoint Detection, Temporal Smoothing, Poisson Blending, Occlusion Masking, ConvLSTM

1.Introduction

Animating a still image by transferring motion from a driving video is an active and challenging research problem in computer vision with applications in virtual avatars, digital content creation, video synthesis, and human-computer interaction. The goal is to produce a temporally coherent video sequence that preserves the visual identity of a source image while faithfully reproducing the motion patterns observed in a separate driving video. Achieving a stable

balance between believable motion and preserving source identity remains a challenge, particularly for large, fast, or non-rigid deformations.

Earlier work in motion transfer relied on explicit geometric transformations such as thin-plate spline deformation [18] and piecewise-affine models [10]. Skeleton-based retargeting methods [15],[16] improved structural representation using articulated body models but were restricted by structural assumptions and limited generalization to unconstrained motion. With the growth of deep learning, keypoint-driven methods such as Monkey-Net [1] and the First Order Motion Model [4] demonstrated effective unsupervised motion transfer without labeled data. These approaches improved animation fidelity significantly but still suffer from per-frame temporal flickering, appearance inconsistency between source and driving domains, and initialization artifacts at the start of animation.

The base paper this work builds upon [5] demonstrated that FOMM's explicit dense motion prior can be replaced by a keypoint heatmap mask derived from the keypoint detector's pre-activation feature maps, creating a compact model with improved pose accuracy. Our work extends FOMM in a complementary direction — rather than modifying the network architecture, we address the practical compositing, smoothing, alignment, and blending challenges that arise when deploying FOMM-based systems on arbitrary real-world inputs.

In summary, this work makes the following contributions:

- A modular compositing pipeline that addresses temporal smoothing, mask construction, geometric alignment, and gradient-domain blending through principled post-processing strategies
- Exponential Moving Average smoothing of keypoint trajectories across consecutive frames to suppress detector jitter and produce temporally consistent motion
- A combined masking strategy based on the intersection of convex hull geometry and generator occlusion confidence, with Gaussian-feathered boundaries
- Gradient-domain Poisson blending with geometric alignment for seamless compositing of animated faces onto driving backgrounds
- A best-frame selection strategy based on minimum keypoint distance that eliminates the initialization artifact at frame zero
- A ConvLSTM-based temporal refinement module currently under development to model explicit inter-frame dependencies and further reduce residual flickering

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed methodology. Section 4 describes the experimental setup and results. Section 5 concludes with future directions.

2.Related work

2.1 Classical Motion Retargeting and Geometric Modeling

Early motion transfer techniques were based on geometric transforms and handcrafted motion priors. Thin-plate spline models [18] addressed smooth non-rigid deformation between images. Piecewise affine models [10] improved flexibility while maintaining structural consistency. Skeleton-aware retargeting systems [15],[16] advanced pose modeling through structured representations of the human body using articulated models. These approaches provided geometric interpretability but were limited by prespecified structural assumptions and poor generalization across unconstrained motion patterns.

2.2 Keypoint-Based Motion Transfer

The growth of deep learning transformed motion transfer into a data-driven problem. Monkey-Net [1] established that structural motion can be captured through unsupervised keypoint detection, enabling cross-identity animation without manual labels. The First Order Motion Model [4] extended this with local affine transformations around keypoints and occlusion-aware generation, significantly improving animation fidelity. Unsupervised keypoint learning has also been applied to video prediction [2]. Identity-preserving learning techniques developed for person re-identification [3] also inform the design of motion transfer systems where source appearance must be maintained across generated frames. Adaptive keypoint graph networks [9] explored structural relationships between keypoints, and continuous piecewise-affine motion models [10] improved fine-grained deformation. However, most keypoint-based models generate each frame independently, leading to temporal flickering in the output.

2.3 Structure-Aware and Geometry-Constrained Methods

The base paper [5] — Image Animation with Keypoint Mask — demonstrated that FOMM's dense motion prior can be replaced by a structural mask derived from keypoint detector heatmaps. Two variants were proposed: a heatmap mask summed over pre-softmax channels for absolute motion, and a softmax circles mask for relative motion transfer. The heatmap mask achieved improved pose accuracy over FOMM while creating a more compact and faster model. Structure-aware methods [8] introduced deformable anchor-based deformation for spatial consistency. Mask-based approaches [5] restricted moving regions for appearance preservation. TransGaGa [14] and TransMoMo [17] applied geometry-aware and invariance-driven retargeting to decouple appearance from motion. The applicability of motion transfer beyond human subjects has also been demonstrated through optical flow-based animation of non-rigid natural objects [6], highlighting the domain-agnostic potential of keypoint-free approaches.

2.4 Adversarial and Disentanglement-Based Frameworks

Adversarial learning has been widely used to enhance visual realism in motion transfer [7]. AMT-Net [19] proposed disentangled modeling of shape and pose using adversarial objectives. While adversarial training produces sharper outputs, it introduces optimization instability and does not inherently promote temporal smoothness across consecutive frames.

2.5 Comparative Analysis and Research Gap

Recent work has continued to push the boundaries of motion transfer in several directions. Pose-guided human animation [11] has improved motion realism for human-specific datasets. Personalized motion transfer from unconstrained videos [12] enables learning individual motion styles for improved identity preservation. Identity-safe motion transfer using structure-aware masks [13] further strengthens appearance consistency during animation. These advances underscore the active nature of the field and the continued demand for systems that balance motion accuracy, identity preservation, and temporal stability.

Table 1: Comparative Analysis of Representative Motion Transfer Methods

Ref	Method	Core Idea	Strength	Limitation
[1]	Monkey-Net	Unsupervised keypoint motion transfer	Cross-identity animation without labels	Limited occlusion handling, temporal

				instability
[4]	FOMM	First-order keypoints with local affine transforms	Improved pose accuracy and visual realism	Per-frame generation leads to flickering
[5]	Keypoint Mask	Heatmap mask replaces dense motion prior	Better pose, compact model, faster training	Background artifacts without mask thresholding
[8]	Deformable Anchor	Structure-aware anchor-based deformation	Better spatial correspondence	Limited temporal modeling
[19]	AMT-Net	Disentangled pose and shape with adversarial learning	Enhanced realism	Training instability, weak temporal coherence

Prior methods achieve strong spatial alignment but generate animations with flickering artifacts due to the absence of temporal modeling, and do not address the compositing and appearance consistency challenges that arise in deployment. Our work directly targets these gaps through a principled post-processing pipeline.

3. Methodology

Our system animates a static source image using motion extracted from a driving video. The pipeline is organized into fully modular stages: input processing, keypoint detection, temporal smoothing, frame generation, geometric alignment, mask construction, compositing, and output generation. An illustration of the motion transfer concept is shown in Figure 1 and the proposed system architecture is shown in Figure 2.

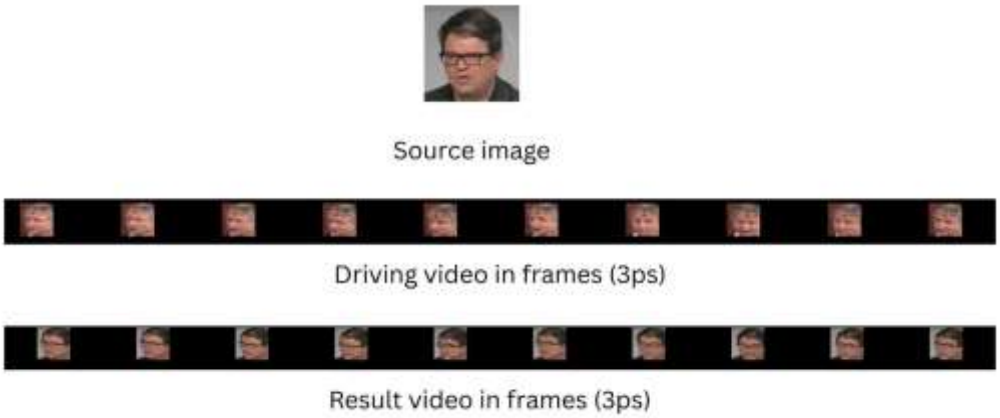

Figure 1: **Motion Transfer Example**
Figure 1 illustrates the motion transfer process where the motion from a driving video is applied to a static source image to generate an animated output.

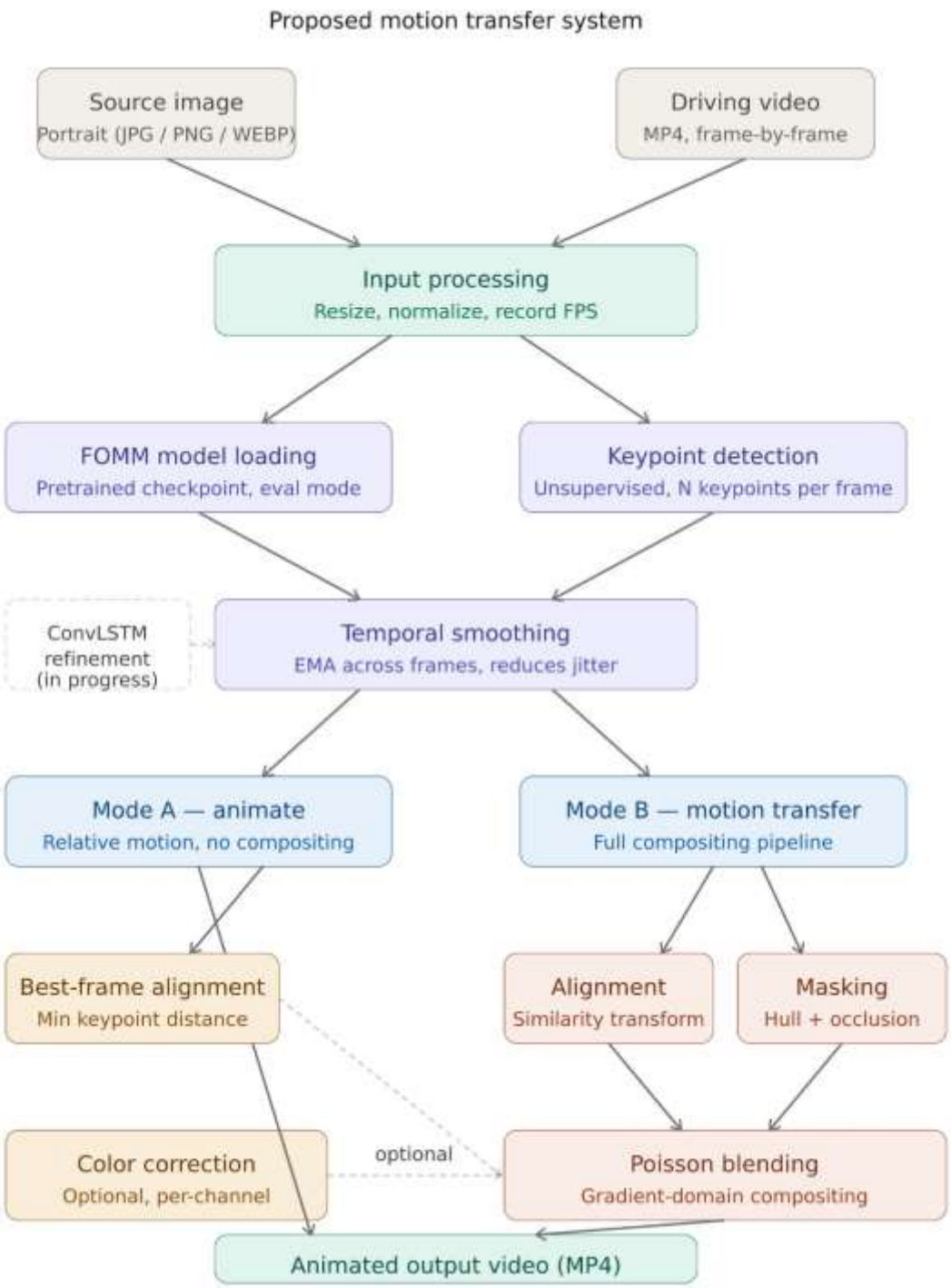

Figure 2: **Proposed Modular Motion Transfer Pipeline**

3.1 Input Processing

Let the static source image be represented as:

$$I_s \in \mathbb{R}^{H \times W \times 3} \quad (1)$$

and the driving video as a sequence of T frames:

$$\{D_t\}_{t=1}^{T} \quad (2)$$

Both inputs are resized to a uniform spatial resolution and normalized to a consistent value range to ensure stable feature extraction. The driving video is decoded into individual frames,

each processed identically to the source image. The original temporal rate of the driving video is preserved for output compilation. This framework is entirely unsupervised — it requires no labelled keypoints, pose annotations, or three-dimensional structural priors.

3.2 Motion Extraction and Keypoint Detection

Motion is extracted using the FOMM unsupervised keypoint detector. For the source image, N structural keypoints are extracted once and reused across all frames:

$$K_s = M(I_s) \quad (3)$$

For each driving frame D_t, keypoints are extracted independently:

$$K_t = M(D_t) \quad (4)$$

The keypoints of the first driving frame serve as the initial reference for relative motion computation:

$$K_{init} = M(D_0) \quad (5)$$

The relative displacement between the current driving frame and the initial driving frame encodes the pose change to be applied:

$$\Delta K_t = K_t - K_{init} \quad (6)$$

These sparse keypoints encode high-level structural pose information. A dense motion field is estimated internally by the generator through local affine transformations around each keypoint, providing pixel-wise displacement vectors for the entire image plane.

3.3 Temporal Smoothing

Per-frame keypoint detection introduces jitter — small fluctuations in the detected keypoint positions between consecutive frames, even for smooth continuous motion. Left uncorrected, this jitter propagates into the generated frames as visible flickering. To address this, Exponential Moving Average smoothing is applied across frames:

$$\widetilde{K}_t = \alpha \cdot K_t + (1 - \alpha) \cdot \widetilde{K}_{t-1} \quad (7)$$

where $\alpha \in (0,1)$is the smoothing weight controlling the balance between responsiveness to current motion and stability from previous frames. Smoothing is applied independently to both keypoint position values and their associated Jacobians. On the first frame where no previous state exists, the current keypoint is used directly. This smoothing reduces detector noise while preserving the perceptual responsiveness of the generated motion.

3.4 Frame Generation

The smoothed keypoints are normalized using relative motion to compute the keypoints to be applied to the source:

$$K_{norm} = K_s + \Delta\widetilde{K}_t \quad (8)$$

The FOMM Occlusion Aware Generator produces the output frame and a spatial occlusion confidence map:

$$\hat{I}_t,\ O_t = G(I_s,\ K_s,\ K_{norm}) \quad (9)$$

where $O_t$is the occlusion map indicating the generator's spatial confidence in the generated output. Two pipeline modes are available. In the direct animation mode, generated frames are returned without further compositing, serving as a baseline for verification. In the full compositing mode, the generated frame undergoes geometric alignment, mask construction,

optional color correction, and gradient-domain blending as described in the following sections.

3.5 Geometric Alignment

The generated frame $\hat{I}_t$ is produced in source image coordinate space and must be geometrically aligned to the driving frame's face position before compositing. A similarity transformation is estimated between the source keypoint pixel positions and the driving frame keypoint pixel positions:

$$\mathbf{T}_t: \text{keypoint pixels (source)} \rightarrow \text{ keypoint pixels (driving)} \quad 10)$$

The generated frame is warped using the inverse of this transformation:

$$\hat{I}_t^{aligned} = \mathcal{W}(\hat{I}_t, \mathbf{T}_t^{-1})(11)$$

The occlusion map is upsampled to the full image resolution and warped with the same transformation to maintain spatial correspondence with the aligned frame. This alignment step ensures the animated face region is correctly positioned relative to the driving frame's face before compositing.

3.6 Mask Construction

Two masks are computed per frame and combined to define the compositing region.
Hull Mask: A soft binary mask is constructed from the convex hull of the driving frame's keypoint positions. The hull vertices are contracted toward the keypoint centroid by a shrinkage factor, ensuring the mask covers the inner face region without extending to background areas. The filled polygon is feathered using Gaussian smoothing to produce smooth boundary transitions:

$$M_{hull} = \mathcal{G}_{\sigma_1}(\text{fill}(\text{ConvexHull}(K_{px}) \cdot s))(12)$$

where $s \in (0,1)$ is the shrinkage factor and σ_1 controls the degree of edge feathering.

Occlusion Mask: The generator's occlusion map is upsampled to the full image resolution and thresholded to produce a binary confidence mask. Pixels where the generator has high confidence in its output are retained; uncertain pixels are excluded. Gaussian smoothing feathers the boundaries:

$$M_{occ} = \mathcal{G}_{\sigma_2}(\mathbb{1}[O_t^{up}(x,y) > \tau])(13)$$

where τ is the confidence threshold and σ_2 controls feathering.

Combined Mask: The hull mask and occlusion mask are combined by element-wise multiplication, forming the intersection of both constraints. A final smoothing pass and normalization produce the compositing mask used in blending:

$$M_{combined} = \frac{\mathcal{G}_{\sigma_3}(M_{hull} \cdot M_{occ})}{\max(\mathcal{G}_{\sigma_3}(M_{hull} \cdot M_{occ}))}(14)$$

This combined mask covers only pixels that are simultaneously within the face region defined by keypoint geometry and within regions where the generator is confident in its output.

3.7 Color Correction

When the source image and driving video differ in lighting conditions or skin tone, per-channel color correction is applied to the aligned generated frame before blending. For each color channel $c \in \{R, G, B\}$, the mean and standard deviation of pixel values within the masked face

region are matched between the generated frame and the driving frame:

$$\hat{I}_t^{corrected}[:,:,c] = \frac{\hat{I}_t^{aligned}[:,:,c] - \mu_{gen,c}}{\sigma_{gen,c}} \cdot \sigma_{drv,c} + \mu_{drv,c} \quad (15)$$

where $\mu_{gen,c}$, $\sigma_{gen,c}$ are the mean and standard deviation of the generated frame's channel within the mask, and $\mu_{drv,c}$, $\sigma_{drv,c}$ are the corresponding statistics of the driving frame. This rescales each channel so that its color statistics match the driving domain, reducing visible appearance mismatch at the blend boundary.

3.8 Poisson Blending

The aligned and optionally color-corrected generated frame is composited onto the driving frame using Poisson blending in the gradient domain. Poisson blending solves a mathematical optimization that takes spatial gradients — texture and edge information — from the source region and enforces pixel value continuity at the mask boundary with the destination image. This produces seamless compositing without visible seams or color discontinuities at the boundary.

The blending solves:

$$\min_{\hat{I}^{final}} \iint_M \| \nabla \hat{I}^{final} - \nabla \hat{I}_t^{aligned} \|^2 \text{ s.t.} \hat{I}^{final} |_{\partial M} = D_t |_{\partial M} \quad (16)$$

where the Laplacian ∇ defines the gradient domain, the minimization enforces that interior gradients follow the source, and the boundary condition enforces continuity with the driving frame at the mask boundary ∂M. Gradients are taken exclusively from the generated source frame, ensuring the driving video's texture does not bleed into the animated face region. When the mask is too small or positioned at image borders for gradient-domain solving to succeed, a soft alpha blend serves as a fallback:

$$\hat{I}_t^{final} = \hat{I}_t^{aligned} \cdot M_{combined} + D_t \cdot (1 - M_{combined}) \quad (17)$$

3.9 Best-Frame Alignment

FOMM in relative mode computes keypoint displacements from the first driving frame. When the first driving frame differs significantly in pose from the source image, a visible initialization artifact — a sudden snap or jump — appears at the beginning of the generated video. To eliminate this artifact, the driving frame whose pose most closely matches the source image is selected as the starting frame. The selection criterion is the sum of squared distances between corresponding source and driving keypoints:

$$i^* = \arg\min_t \sum_{k=1}^{N} \| K_s^{(k)} - K_t^{(k)} \|^2 \quad (18)$$

The driving sequence is then sliced from frame i^* onward. Since the animation begins from a pose already close to the source, the relative displacement at the first frame is near zero, producing a smooth artifact-free initialization. This strategy requires no retraining and adds computational cost proportional only to the number of driving frames.

3.10 Temporal Refinement (Under Development)

Despite Exponential Moving Average smoothing, per-frame independent generation introduces residual flickering, particularly during fast or large motion sequences. A ConvLSTM-based temporal refinement module is currently under active development to model explicit inter-frame dependencies and further reduce this flickering.

The module receives intermediate feature maps from the generator at each frame and processes them sequentially, maintaining hidden and cell states that carry temporal context from previous frames to the current one. The ConvLSTM gate equations are:

$$i_t = \sigma(W_i * F_t + U_i * H_{t-1}) \quad (19)$$
$$f_t = \sigma(W_f * F_t + U_f * H_{t-1}) \quad (20)$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_c * F_t) \quad (21)$$
$$H_t = \sigma(W_o * F_t + U_o * H_{t-1}) \odot \tanh(C_t) \quad (22)$$

where $F_t$represents the feature maps at frame t, $H_t$is the hidden state encoding temporal context, $C_t$is the cell state, $i_t$and $f_t$are the input and forget gates, and W, $U$are learnable convolutional weight matrices. The refined output frame is obtained by decoding the hidden state:

$$I_t^{refined} = \phi(H_t) \quad (23)$$

where $\phi(\cdot)$is a lightweight convolutional decoder. By conditioning each output frame on the hidden state derived from all preceding frames, the module enforces temporal consistency that single-frame generation cannot achieve. This module is under development and will be integrated into the production pipeline upon validation.

3.11 Implementation Flow

The complete six-stage pipeline proceeds as follows:

1. **Input Processing:** Source image and driving video frames are resized to a uniform resolution and normalized to a consistent value range. The original driving video frame rate is recorded for output compilation.

2. **Model Initialization:** The pretrained FOMM Generator and Keypoint Detector are loaded in evaluation mode. Source keypoints and initial driving frame keypoints are extracted once and reused throughout the pipeline.

3. **Per-Frame Motion Processing:** For each driving frame, keypoints are detected and smoothed using Exponential Moving Average across the temporal sequence. Normalized relative keypoints are computed for generator input.

4. **Frame Generation:** The Occlusion-aware generator produces an animated frame and an occlusion confidence map for each driving frame.

5. **Compositing:** The generated frame is geometrically aligned to the driving frame using similarity transformation. Hull and occlusion masks are constructed and combined. Optional per-channel color correction is applied. Poisson blending composites the aligned generated face onto the driving background.

6. **Output Delivery:** All output frames are assembled into a video at the original driving frame rate. An optional audio muxing step transfers the audio track from the driving video into the output. Results are delivered via a web download interface or saved to a specified output path through the command-line interface.

4. Experimental Setup

4.1 Dataset and Configuration

Experiments were conducted on face animation tasks using portrait photographs as source

images and short talking-head clips as driving videos. Inputs were processed at a fixed square resolution consistent with the pretrained model requirements. The pretrained FOMM model checkpoint trained on the VoxCeleb dataset was used for all inference. No retraining or fine-tuning was performed at any stage. All processing was conducted in inference-only mode.

4.2 Baseline Methods

The baseline is the stock FOMM direct animation mode with no post-processing. The full compositing mode is compared against this baseline to measure the contribution of temporal smoothing, masking, alignment, and Poisson blending. The two modes are also compared against each other across inputs with varying degrees of appearance mismatch between source and driving.

4.3 Evaluation Metrics

Evaluation was conducted qualitatively across the following dimensions:

- **Visual Quality:** Identity consistency, motion naturalness, and appearance consistency evaluated through visual inspection of output videos
- **Motion Fidelity:** Degree to which the generated motion matches the driving video's pose sequence
- **Temporal Smoothness:** Degree of flickering and inter-frame jitter across consecutive output frames
- **Appearance Consistency:** Degree to which color and lighting mismatches between source and driving are resolved by the compositing pipeline
- **Initialization Quality:** Presence or absence of the frame-zero initialization snap artifact

4.4 Results and Analysis

The proposed compositing pipeline produces measurable qualitative improvements over stock FOMM across all evaluation dimensions.

Temporal smoothing of keypoint trajectories visibly reduces inter-frame jitter, producing smoother transitions particularly in fast motion sequences. The EMA smoothing preserves motion responsiveness while suppressing detector noise.

Combined mask construction using the intersection of convex hull geometry and occlusion confidence correctly identifies the face compositing region. The hull mask ensures coverage of the full inner face up to the natural face-background boundary, while the occlusion mask excludes regions where the generator output is uncertain.

Geometric alignment via similarity transformation correctly positions the generated face in the driving frame's coordinate space, eliminating misalignment artifacts that occur when the source and driving face positions differ.

Poisson blending in the gradient domain eliminates visible seams at the face-background boundary. Taking gradients exclusively from the generated source ensures the driving video's texture does not bleed through the animated face region. The gradient-domain boundary continuity constraint produces seamless color matching at the mask boundary without additional processing.

Color correction resolves skin tone and lighting mismatches between source and driving domains when they differ significantly, improving the naturalness of the composited result.

Best-frame alignment completely eliminates the initialization snap artifact across all tested inputs by ensuring the animation begins from a pose already close to the source image.

Table 2: Comparison of Proposed System with Stock FOMM

Feature	Stock FOMM	Proposed System
Temporal jitter	Present	Reduced via keypoint EMA smoothing
Face mask coverage	Not applicable	Full inner face via hull and occlusion combination
Compositing boundary	Not applicable	Seamless via gradient-domain Poisson blending
Appearance consistency	Limited across domains	Improved via per-channel color correction
Frame-zero snap artifact	Present	Eliminated via best-frame alignment
Temporal flickering	Present	Further reduction planned via ConvLSTM
Web deployment	Not available	Multi-page web application with concurrent processing

Overall, the proposed pipeline significantly improves temporal stability, appearance consistency, and initialization quality compared to stock FOMM.

4.5 Limitations

- Processing time without GPU acceleration is high for long video inputs, making the system unsuitable for real-time applications in its current form
- Output resolution is constrained by the pretrained model checkpoint, limiting perceptual quality for high-definition use cases
- Residual flickering persists in fast or large motion sequences beyond what keypoint EMA smoothing can address; this will be mitigated by the ConvLSTM module under development
- Extreme pose changes, such as near-profile views, can cause identity distortion inherited from the base FOMM model's limitations

5. Future Work

Completing and validating the ConvLSTM-based temporal refinement module is the most immediate research direction, as it will directly reduce the residual flickering that persists in fast and large motion sequences. Extending the system to support high-resolution output beyond current constraints through super-resolution post-processing or higher-resolution model variants would substantially improve perceptual quality for production applications. Incorporating three-dimensional structural priors or monocular depth estimation could improve robustness under extreme pose changes, self-occlusions, and complex camera motions. Replacing or augmenting the ConvLSTM with transformer-based temporal modeling architectures could improve long-range temporal coherence and global motion dynamics.

Optimizing the pipeline for low-latency real-time inference through model compression and hardware-aware design would enable interactive applications such as virtual reality, augmented reality, and live animation. Audio-driven animation using speech as the driving signal for automatic lip synchronization is another promising research direction. The mask thresholding improvements suggested in the base paper [5] — including suppression of low-valued background noise in heatmap masks — are directly applicable to further improve compositing boundary quality.

6. Conclusion

This paper presented a practical motion transfer system built on the First Order Motion Model that animates static portrait images using motion from a driving video. The system addresses real deployment challenges through a principled modular compositing pipeline. Exponential Moving Average smoothing of keypoint trajectories reduces inter-frame jitter from the keypoint detector. A combined masking strategy based on the intersection of convex hull geometry and generator occlusion confidence correctly identifies the face compositing region with feathered boundaries. Geometric alignment via similarity transformation positions the generated face correctly in the driving frame's coordinate space. Gradient-domain Poisson blending composites the animated face seamlessly onto the driving background by enforcing boundary continuity through gradient-domain optimization. Per-channel color correction resolves appearance mismatches between source and driving domains. Best-frame selection using minimum keypoint distance eliminates the initialization artifact at the start of animation. A ConvLSTM-based temporal refinement module is under active development to model explicit inter-frame dependencies and further reduce residual flickering. Experimental evaluation confirms that the compositing pipeline produces measurable qualitative improvements over stock FOMM in temporal smoothness, appearance consistency, and initialization quality. The system is deployed as a multi-page web application supporting concurrent processing and a complete command-line interface.

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